SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	M. Yasodha krishna	**Reg. No.:**	192472100
Section / Batch:	CSE-AI	**Date:**	16-7-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	100 Marks	

SECTION A – CONCEPT MAPPING

Question-1

consider the concept map: Algorithm → Input size(n) → Growth rate → Efficiency. Explain how each node influence the next as n increa-ses. Extend the map by linking asymptotic notations and scalability as supporting nodes. Show the relationship between input growth and algorithm performance

Question-2

conside the concept map: Algorithm → Growth Function → Big-Theta notations → Tight Bound → Exact order of growth. Explain how the concept of tight bound emerges when upper and lower bounds meet. Extend the map by adding links from Big-O and Big-omega toward Big-Theta to show the relationship.

Code of Conduct 192472100

I M.Yasodha Krishna (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.Yasod

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1	4	4	3	3	2	43
Q2	4	4	3	2	2	40
					Overall Total (100)	83

Faculty In-charge	Course Coordinator	Head of Department
Signature:	Signature:	Signature:
Date:	Date:	Date:

CSA0619

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CO1 - AT2 - Concept Mapping

Relationship Between Input Growth and Algorithm Performance

Algorithm
A step by step Procedure to solve a Problem

Influence
The choice of algorithm determines how the Problem is approached and what operations are performed

Input Size(n)
Number of elements or amount of data Processed by the Algorithm

Influence
As n increases, more data is processed, affecting the resources required

Growth Rate
How the Time or space required by the algorithm increases as n increases

Influence
Determines how rapidly the resources grow with n.

Efficiency
Ability of the Algorithm to solve the Problem using time and space effectively

Influence
lower growth rate means better efficiency and faster performance for large n.

Asymptotic Notations
mathematical tools to describe growth rates for larger, ignoring constants and low order terms.

Examples: O( ), Ω( ), θ( ), o( ) ω( )

Scalability
Ability of Algorithm to maintain good performance as input size grows very large

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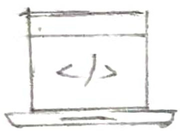
Concept Map: From Algorithm to Exact Order of Growth

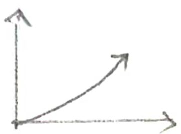
Algorithm



A step by step procedure to solve a problem

→

Growth Function



A function $T(n)$ that represents time or space required by the Algorithm as a funct of input size n

→

Big-theta notation

(Θ)

(Θ)

describe functions that grows at the same rate up to constant factors

→

Tight Bound

When upper and lower bounds meet

→

Exact order of Growth

The precise rate at which the algorithm resource usage grows as $n \to \infty$

Big-O notation (O)
(Upper Bound)

$T(n)$ is asymptotically less than or equal to a function $g(n)$

$T(n) = O(g(n))$

Provides upper Bound →

Big-Theta Notation (Θ)
(Tight Bound)

$T(n) = \Theta(g(n))$

$\exists$ positive constants c_1, c_2 & c_0 such that for all $n \geq n_0$

$0 \leq c_1 g(n) \leq T(n) \leq c_2 g(n)$

↑ Big-Theta represents tight bound.

← Provides lower Bound

Big-Omega notation (Ω)
(lower Bound)

$T(n)$ is asymptotically greater than or equal to a function $g(n)$

$T(n) = \Omega(g(n))$